WORKED SOLUTION

STEP III 2020, Question 6

Pure • Polar Curves and Area

STEP III 2020, Question 6

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

All three parts are built on the same function $\cos\theta + \sqrt{\cos 2\theta}$. The equation of C_2 is a quadratic in r, and its two roots are $\cos\theta \pm \sqrt{\cos 2\theta}$. So C_2 is C_1 (the larger root) together with a second, inner branch (the smaller root), and the two branches meet where the discriminant $\cos 2\theta$ vanishes. With that in place, the area between the branches collapses to a single integral, because $r_+^2 - r_-^2$ factorises.

Solution

Part (i)

Let $y = \cos x + \sqrt{\cos 2x}$. This is real exactly when $\cos 2x \geq 0$, which on the given interval is the whole of $[-\frac{1}{4}\pi, \frac{1}{4}\pi]$. The function is even, so the graph is symmetric about the y-axis.

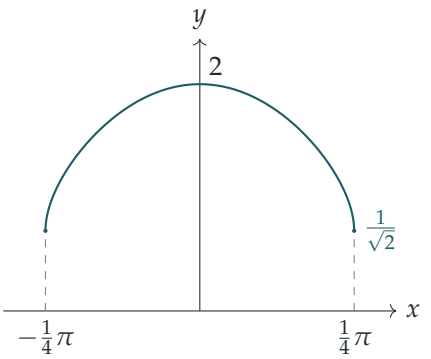
Values. At $x = 0$, $y = 2$. At $x = \pm\frac{1}{4}\pi$, $\cos 2x = 0$ and $y = \cos\frac{1}{4}\pi = \frac{1}{\sqrt{2}}$.

Gradient. For $0 < x < \frac{1}{4}\pi$,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\sin x - \frac{\sin 2x}{\sqrt{\cos 2x}} < 0,$$

since both terms are negative there. So y decreases from 2 to $\frac{1}{\sqrt{2}}$ as x goes from 0 to $\frac{1}{4}\pi$, and the maximum is at $(0,2)$. As $x \to \frac{1}{4}\pi^-$, $\sqrt{\cos 2x} \to 0^+$ while $\sin 2x \to 1$, so $\mathrm{d}y/\mathrm{d}x \to -\infty$: the curve meets the end points with *vertical* tangents.

Near $x = 0$. Using $\cos x \approx 1 - \frac{1}{2}x^2$ and $\sqrt{\cos 2x} \approx \sqrt{1 - 2x^2} \approx 1 - x^2$, we get $y \approx 2 - \frac{3}{2}x^2$, a smooth maximum.



Part (ii)

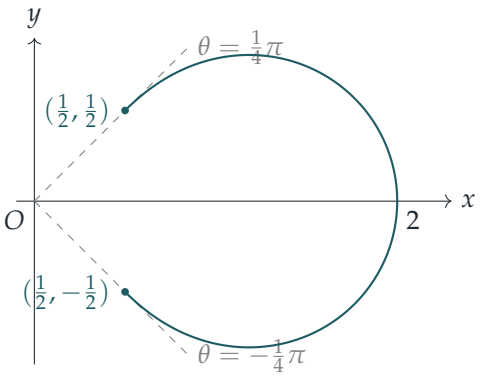
Now the same function gives the distance r from the origin in direction θ. The end points of C_1 are at $r = \frac{1}{\sqrt{2}}$, $\theta = \pm\frac{1}{4}\pi$, which are the Cartesian points $(\frac{1}{2}, \pm\frac{1}{2})$. At $\theta = 0$ the curve is at $(2, 0)$, the furthest point from O. As r is even in θ, C_1 is symmetric in the initial line.

Ask yourself What does an infinite $\mathrm{d}r/\mathrm{d}\theta$ mean for a polar curve?

From part (i), $\mathrm{d}r/\mathrm{d}\theta \to -\infty$ as $\theta \to \frac{1}{4}\pi^-$. Near that end, r changes a lot while θ hardly changes at all, so the curve is moving along the radial line $\theta = \frac{1}{4}\pi$ (the line $y = x$). Since r is decreasing, it arrives at $(\frac{1}{2}, \frac{1}{2})$ heading *towards* the origin along $y = x$. In particular, the height $y = r\sin\theta$ must reach a maximum before the end point. Indeed

$$\frac{\mathrm{d}y}{\mathrm{d}\theta} = \frac{\mathrm{d}r}{\mathrm{d}\theta}\sin\theta + r\cos\theta$$

is positive near $\theta = 0$ and tends to $-\infty$ as $\theta \to \frac{1}{4}\pi$. (The maximum height is about 0.81, at $\theta \approx 34°$.) So C_1 is an arch through $(2, 0)$ that bulges above $y = \frac{1}{2}$ and comes back down to $(\frac{1}{2}, \frac{1}{2})$, touching the line $y = x$ there, with a mirror image below the axis.



Part (iii)

The two branches. Treat the equation as a quadratic in r:

$$r^2 - 2r\cos\theta + \sin^2\theta = 0 \iff r = \cos\theta \pm \sqrt{\cos^2\theta - \sin^2\theta} = \cos\theta \pm \sqrt{\cos 2\theta}.$$

Write $r_\pm = \cos\theta \pm \sqrt{\cos 2\theta}$. The outer branch r_+ is exactly C_1. The inner branch satisfies $r_- \geq 0$, because $\cos^2\theta \geq \cos^2\theta - \sin^2\theta = \cos 2\theta$, with equality only when $\sin\theta = 0$. So r_- is a genuine polar curve, and it reaches the origin only at $\theta = 0$.

At $\theta = \pm\frac{1}{4}\pi$. Here $\cos 2\theta = 0$, so the quadratic has a repeated root:

$$r = \cos\tfrac{1}{4}\pi = \frac{1}{\sqrt{2}}.$$

Both branches pass through $(\frac{1}{2}, \pm\frac{1}{2})$.

Small r. If r is small, then r^2 is negligible next to $2r\cos\theta$, and θ must be small too, since $\sin^2\theta = 2r\cos\theta - r^2$ is small. Then $2r\cos\theta \approx \sin^2\theta$, and using $\cos\theta \approx 1$, $\sin\theta \approx \theta$,

$$r \approx \frac{\sin^2\theta}{2\cos\theta} \approx \tfrac{1}{2}\theta^2,$$

as required. (Equivalently, $r_- \approx (1 - \frac{1}{2}\theta^2) - (1 - \theta^2) = \frac{1}{2}\theta^2$.)

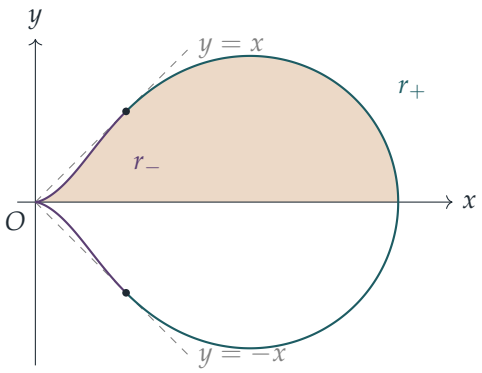
Behaviour near $r = 0$. Here $x = r\cos\theta \approx \frac{1}{2}\theta^2$ and $y = r\sin\theta \approx \frac{1}{2}\theta^3$, so $y^2 \approx 2x^3$. The inner branch therefore comes into the origin tangent to the initial line from the side $x > 0$, from above and below, forming a *cusp* at O that points left.

Behaviour near $\theta = \pm\frac{1}{4}\pi$. Differentiating the equation implicitly,

$$(2r - 2\cos\theta)\,\mathrm{d}r + (2r\sin\theta + 2\sin\theta\cos\theta)\,\mathrm{d}\theta = 0 \quad\Longrightarrow\quad \frac{\mathrm{d}\theta}{\mathrm{d}r} = \frac{\cos\theta - r}{\sin\theta\,(r + \cos\theta)}.$$

At $r = \frac{1}{\sqrt{2}}$, $\theta = \frac{1}{4}\pi$ the numerator is 0 and the denominator is not, so $\mathrm{d}\theta/\mathrm{d}r = 0$. Also $\theta \leq \frac{1}{4}\pi$ on the whole curve. So θ has a maximum there, and the curve is tangent to the radial line $y = x$. The outer branch comes in along $y = x$ towards O, the inner branch leaves along $y = x$ away from O, and the two join smoothly at $(\frac{1}{2}, \frac{1}{2})$. There is no corner. The same happens at $(\frac{1}{2}, -\frac{1}{2})$ by symmetry.

So C_2 is a closed curve, symmetric in the initial line, shaped like a teardrop whose point is the cusp at O. Its outer edge is C_1, and its inner edge runs from $(\frac{1}{2}, \frac{1}{2})$ down into the cusp at O and out again to $(\frac{1}{2}, -\frac{1}{2})$.



(The plotting uses $\sin\theta = \frac{1}{\sqrt{2}}\sin u$, which makes $\sqrt{\cos 2\theta} = \cos u$. Letting u run once round traces the whole of C_2 without any infinite slopes.)

The area. For $0 \le \theta \le \frac{1}{4}\pi$, the ray at angle θ is inside C_2 exactly for $r_- \le r \le r_+$. So the shaded area is the difference of two polar sectors:

$$\begin{aligned} A &= \int_0^{\pi/4} \tfrac{1}{2}\left(r_+^2 - r_-^2\right) \mathrm{d}\theta = \int_0^{\pi/4} \tfrac{1}{2}(r_+ - r_-)(r_+ + r_-)\,\mathrm{d}\theta \\ &= \int_0^{\pi/4} \tfrac{1}{2}\cdot 2\sqrt{\cos 2\theta}\cdot 2\cos\theta\,\mathrm{d}\theta = 2\int_0^{\pi/4} \cos\theta\sqrt{1-2\sin^2\theta}\,\mathrm{d}\theta. \end{aligned}$$

Substitute $\sqrt{2}\sin\theta = \sin u$, so $\sqrt{2}\cos\theta\,\mathrm{d}\theta = \cos u\,\mathrm{d}u$, and u runs from 0 to $\frac{1}{2}\pi$ as θ runs from 0 to $\frac{1}{4}\pi$:

$$A = 2\int_0^{\pi/2} \cos u\cdot\frac{\cos u}{\sqrt{2}}\,\mathrm{d}u = \sqrt{2}\int_0^{\pi/2}\cos^2 u\,\mathrm{d}u = \sqrt{2}\cdot\frac{\pi}{4} = \frac{\pi}{2\sqrt{2}},$$

which is the required result. (Equivalently, the substitution $s = \sqrt{2}\sin\theta$ turns the integral into $\sqrt{2}\int_0^1 \sqrt{1-s^2}\,\mathrm{d}s$, which is $\sqrt{2}$ times a quarter of the unit disc.)

Checks

- At $\theta = 0$, $r_\pm = 1 \pm 1$, giving the points $(2,0)$ and O, consistent with the sketch.
- At $\theta = 0.01$, $r_- = \cos 0.01 - \sqrt{\cos 0.02} \approx 5.0002\times 10^{-5}$, against $\frac{1}{2}\theta^2 = 5\times 10^{-5}$.
- Numerically, $\pi/(2\sqrt{2}) \approx 1.11072$, and a direct numerical integration of $\frac{1}{2}(r_+^2 - r_-^2)$ over $[0, \frac{1}{4}\pi]$ gives the same value to six decimal places.
- A rough bound: the upper half of C_2 lies inside the rectangle $[0,2]\times[0,0.81]$, of area about 1.62, and it contains the triangle with vertices $(\frac{1}{2},0)$, $(2,0)$, $(\frac{1}{2},\frac{1}{2})$, of area 0.375. The value 1.11 sits sensibly between these.

Common slips

- Sketching C_1 with a corner, or with the curve crossing $y = x$, at $(\frac{1}{2}, \frac{1}{2})$: the infinite $\mathrm{d}r/\mathrm{d}\theta$ means the curve touches the radial line there.
- Drawing a smooth rounded end at O instead of a cusp: $y^2 \approx 2x^3$ is the classic cusp shape.
- Computing $\int \frac{1}{2} r_+^2 \, \mathrm{d}\theta$ alone, which is the area out to C_1 and includes the hollow region inside the inner branch.

Beyond STEP

The curve $y^2 = 2x^3$ near the origin is a semicubical parabola, the standard example of a cusp. Cusps show up whenever a smooth parametrisation has velocity zero at some instant: here $(x, y) \approx (\frac{1}{2}\theta^2, \frac{1}{2}\theta^3)$ has zero derivative at $\theta = 0$. Classifying such singular points of algebraic curves is the start of algebraic geometry. Polar curves and their areas also lead naturally to Green's theorem, which expresses an enclosed area as a line integral round its boundary. For a geometric view of curves like these, try Needham, *Visual Complex Analysis*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.